

Lecture 21-22

Non-Parametric Tests

- Non-parametric $\rightarrow$ we impose fewer assumptions abt. the exact form of the population distribution. $\Rightarrow$ broader applicability, but usually less info. extract from data.
- Type I $\rightarrow$ reject H_0 , when H_0 is true. $\Rightarrow \alpha$
Type II $\rightarrow$ fail to reject H_0 , when H_0 is false $\Rightarrow \beta$
 $n \uparrow \Rightarrow \beta \downarrow \Rightarrow \text{Power} = 1 - \beta \uparrow$
- Sign Test $\rightarrow$ nonparametric alternative to a 1-sample t-test

$H_0: \mu = \mu_0$ each x_i $x_i > \mu_0 \Rightarrow +$
 $x_i < \mu_0 \Rightarrow -$
 $x_i = \mu_0 \Rightarrow \text{discard}$

under H_0 , μ_0 is the population median,
 $P(X > \mu_0) = \frac{1}{2}$ & $P(X < \mu_0) = \frac{1}{2}$

X be the no. of $+$ $\Rightarrow X \sim \text{Bin}(n, \frac{1}{2})$

$$E[X] = \frac{n}{2} \quad \text{Var}(X) = \frac{n}{4} \quad \text{SD}(X) = \frac{\sqrt{n}}{2}$$

$H_1: \mu > \mu_0$, a large no. $+$ signs
 $H_1: \mu < \mu_0$, lesser $+$ signs

$$X \sim \text{Bin}(n, 0.5)$$

$$\begin{array}{l} \text{upper} \Rightarrow p = P(X \geq X_{\text{obs}}) \\ \text{lower} \Rightarrow p = P(X \leq X_{\text{obs}}) \end{array}$$

but when samples are large

$$X \sim N\left(\frac{n}{2}, \frac{n}{4}\right) \Rightarrow Z = \frac{X - n/2}{\sqrt{n}/2}$$

The sign test is naturally a test abt. the population median.

$$P(X \geq k) = 1 - P(X \leq k)$$

$$P(X = k) = \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}$$

Eg. $H_0: \mu = 160$, $H_1: \mu > 160$ (right)
 $\alpha = 0.05$, $n = 19$, $X = 15$

$$X \sim \text{Bin}(19, 0.5)$$

$$p\text{ value} = P(X \geq 15) = \sum_{k=15}^{19} \binom{19}{k} (0.5)^{19}$$

$$p\text{ value} = 0.0096$$

$$0.0096 < 0.05 \Rightarrow p \leq \alpha \Rightarrow \text{reject } H_0$$

- paired sample sign test $\Rightarrow$ pairs (x_i, y_i)

$d_i = x_i - y_i$
 weeks just like
 sign test.

$d_i > 0 \Rightarrow +$
 $d_i < 0 \Rightarrow -$
 $d_i = 0 \Rightarrow \text{ignore}$

Eg. $d_i = \text{bef. accident} - \text{after}$ $d_i > 0$
 $H_0: \mu_1 = \mu_2$ $n = 12$
 $H_1: \mu_1 > \mu_2$ $X = 10$
 $X \sim \text{Bin}(12, 0.5)$

$$p(X \geq 10) = \frac{\binom{12}{10} + \binom{12}{11} + \binom{12}{12}}{2^{12}}$$

$$p = 0.0192 < 0.05 \Rightarrow \text{reject } H_0$$

- disadv. of sign test $\rightarrow$ both +0.001 & +1000 is treated as +, lot of info. is just thrown

- Wilcoxon Signed-Rank Test $\rightarrow$ sign of each diff. + magnitude through rank
 $\downarrow$
 nonparametric alternative to 1-sample t test & paired t-test

$\rightarrow H_0: \mu = \mu_0$, $d_i = x_i - \mu_0$ if paired $d_i = x_i - y_i$

1. remove 0 d_i

2. $|d_i|$

3. rank the diff. smallest to largest

4. in case of ties, assign the avg. rank

5. put original signs to ranks.

C. $T^+ = \text{sum of ranks of + diff.}$
 $T^- = \text{" " " " - " " "}$

$$T^- = \frac{n(n+1)}{2} - T^+$$

for 2 sided test $\Rightarrow T = \min(T^+, T^-)$
 Under H_0 , + & - diff. should be balanced
 signed rank test assumes that distribution
 of the differences is symmetric.

- large sample for signed rank
 $n \geq 15$, $H_0 \Rightarrow E(T^+) = \frac{n(n+1)}{4}$

$$\text{Var}(T^+) = \frac{n(n+1)(2n+1)}{24} \quad \text{SD}(T^+) = \sqrt{\text{Var}(T^+)}$$

$$Z = \frac{T^+ - \frac{n(n+1)}{4}}{\sqrt{\frac{n(n+1)(2n+1)}{24}}}$$

Ex. $H_0: \mu = 98.5$ $H_1: \mu \neq 98.5$ $\alpha = 0.05$
 $n = 14$
 x_i $d_i = x_i - 98.5$ $|d_i|$ $(\cdot \text{sort } (1))$ rank signed rank

$$T^+ = 2 + 8 = 10, \quad \text{total} = \frac{15(16)}{2} = 120$$

2 sided $T^- = 120 - 10 = 110$

$$T = \min(T^+, T^-) = 10$$

$$T_{0.05} = 21$$

$$10 < 21 \Rightarrow \text{reject } H_0$$

Ex. $d_i = \text{before} - \text{after}$ $H_0: \mu_1 = \mu_2$
 $d_i > 0$ (diff. weeks) $H_1: \mu_1 > \mu_2$

$$T^+ = 111 \quad n = 16 \quad E(T^+) = \frac{16(17)}{4} = 68$$

$$\text{Var} = \frac{16(17)(33)}{24} = 374$$

$$Z = \frac{111 - 64}{\sqrt{374}} \approx 19.34$$

$$Z \approx 2.22$$

$$Z_{0.05} = 1.645$$

$$Z > Z_{0.005} \Rightarrow \text{reject } H_0$$

- Ranked Sum U Test / Mann-Whitney / Wilcoxon Rank-sum
sample 1 $\Rightarrow X_1, \dots, X_{n_1}$, sample 2 $\Rightarrow Y_1, \dots, Y_{n_2}$

1. pool the obs.

2. sort from smallest to largest

3. 1, 2, ..., $n_1 + n_2$

4. for tie, assign avg. rank

5. add the rank belonging to each group.

$$W_1 = \text{sum of ranks of sample 1}$$

$$\text{total} = \frac{(n_1 + n_2)(n_1 + n_2 + 1)}{2}$$

$$W_2 = \frac{(n_1 + n_2)(n_1 + n_2 + 1)}{2} - W_1$$

$$U_1 = W_1 - \frac{n_1(n_1 + 1)}{2}$$

$$U_2 = W_2 - \frac{n_2(n_2 + 1)}{2}$$

$$U = \min(U_1, U_2)$$

Small U means, samples are heavily separated in their ordering.

Eg. $n_1 = 9, n_2 = 10 \quad H_0: \mu_1 = \mu_2 \quad H_1: \mu_1 < \mu_2$

$$W_1 = 69$$

$$U_1 = 69 - \frac{(9)(10)}{2} = 69 - 45 = 24$$

$$p = 0.047 < 0.05 \Rightarrow \text{reject } H_0$$

- if samples are more $n_1 > 8$, & $n_2 > 8$
then normal approx.

$$E(U) = \frac{n_1 n_2}{2}$$

$$\text{Var}(U) = \frac{n_1 n_2 (n_1 + n_2 + 1)}{12}$$

Ex. $\mu_U = 45$ $\sigma_U^2 = 150$

$$Z = \frac{24 - 45}{\sqrt{150}} \approx -1.715 \quad p\text{-value} = 0.043$$

$$0.043 < 0.05 \Rightarrow \text{reject } H_0$$

- Kruskal-Wallis Test $\rightarrow$ rank-based extension of the Mann-Whitney test from 2 groups to k groups.

$$\begin{aligned} 2 \text{ sample } t\text{-test} &\leftrightarrow \text{Mann-Whitney} \\ 1\text{-Way ANOVA} &\leftrightarrow \text{Kruskal Wallis} \end{aligned}$$

- Rank correlation test $\rightarrow$ associaⁿ btw. 2 vars. under normality, we might use Pearson correlaⁿ. Non-parametric alternative is Spearman's rank correlaⁿ coefficient r_s .

$$r_s = \frac{1 - \frac{6 \sum_{i=1}^n d_i^2}{n(n^2 - 1)}}{1}$$

$$d_i = \text{rank}(x_i) - \text{rank}(y_i)$$

$$-1 \leq r_s \leq 1$$

$$r_s = 1 \Rightarrow \text{+ve monotonic associa}^n$$

$$r_s = -1 \Rightarrow \text{-ve " "}$$

$$r_s = 0 \Rightarrow \text{little rank/monotonic associa}^n$$

Pearson $\rightarrow$ tells abt. linear relationship

Spearman $\rightarrow$ tells abt. monotonic "

if obs. has tied value, gives them avg. rank

- H_0 : no associaⁿ

H_1 : associaⁿ exists

large samples, distribuⁿ of r_s is normal

$$E(R_s) = 0 \quad \text{Var}(R_s) = \frac{1}{n-1}$$

$$Z = u_s \sqrt{n-1}$$

- Permutaⁿ Testing

$$X = (8, 10, 12), \quad Y = (15, 17, 18)$$

$$H_0: \mu_x = \mu_y, \quad H_1: \mu_x \neq \mu_y$$

$$\bar{X} = \frac{8+10+12}{3} = 10$$

$$\bar{Y} = \frac{15+17+18}{3} = 16.67$$

$$\bar{X} - \bar{Y} = -6.67, \quad \text{Two-sided} \Rightarrow |\bar{X} - \bar{Y}| = 6.67$$

permutaⁿ says that pool all values, then assign 3 values to X & 3 to Y, repeatedly

$$\binom{6}{3} = 20 \text{ assignments}$$

$$p = \frac{2}{20} = 0.1$$

$$p = \frac{\# \{ \text{permuta}^n \text{ at least as extreme} \}}{\text{total permuta}^n} \quad \begin{matrix} 0.1 < 0.05 \\ \text{reject} \\ H_0 \end{matrix}$$

$$p < \alpha \Rightarrow \text{reject } H_0$$

- MC permutaⁿ test $\Rightarrow$ earlier it was 20 permutaⁿ but $n_1 = n_2 = 50$ - then $\binom{100}{50}$

$\therefore$

$$p \approx \frac{\text{no. of extreme shuffled stats}}{\text{no. of shuffles}}$$

Eg. $n = 20,000$, for hundreds of million of

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

$$\text{at worst } p = 0.5$$

$$SE = 0.00354$$

at 95% error margin

$$\times 1.96 = 0.0069$$

$\therefore$ bias can be much more dangerous than sampling variance,